SRI VASAVI ENGINEERINIG COLLEGE (AUTONOMOUS)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Approved by AICTE,Permanently Affiliated to JNTUK,Kakinada

# SCUD

Volume -10 | Issue -1

# ARSENAL

20+

**Intresting Articels, Theories Snippets , Dept. Galleryv Photography and Paintings.**

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*(18A81A05I5)*

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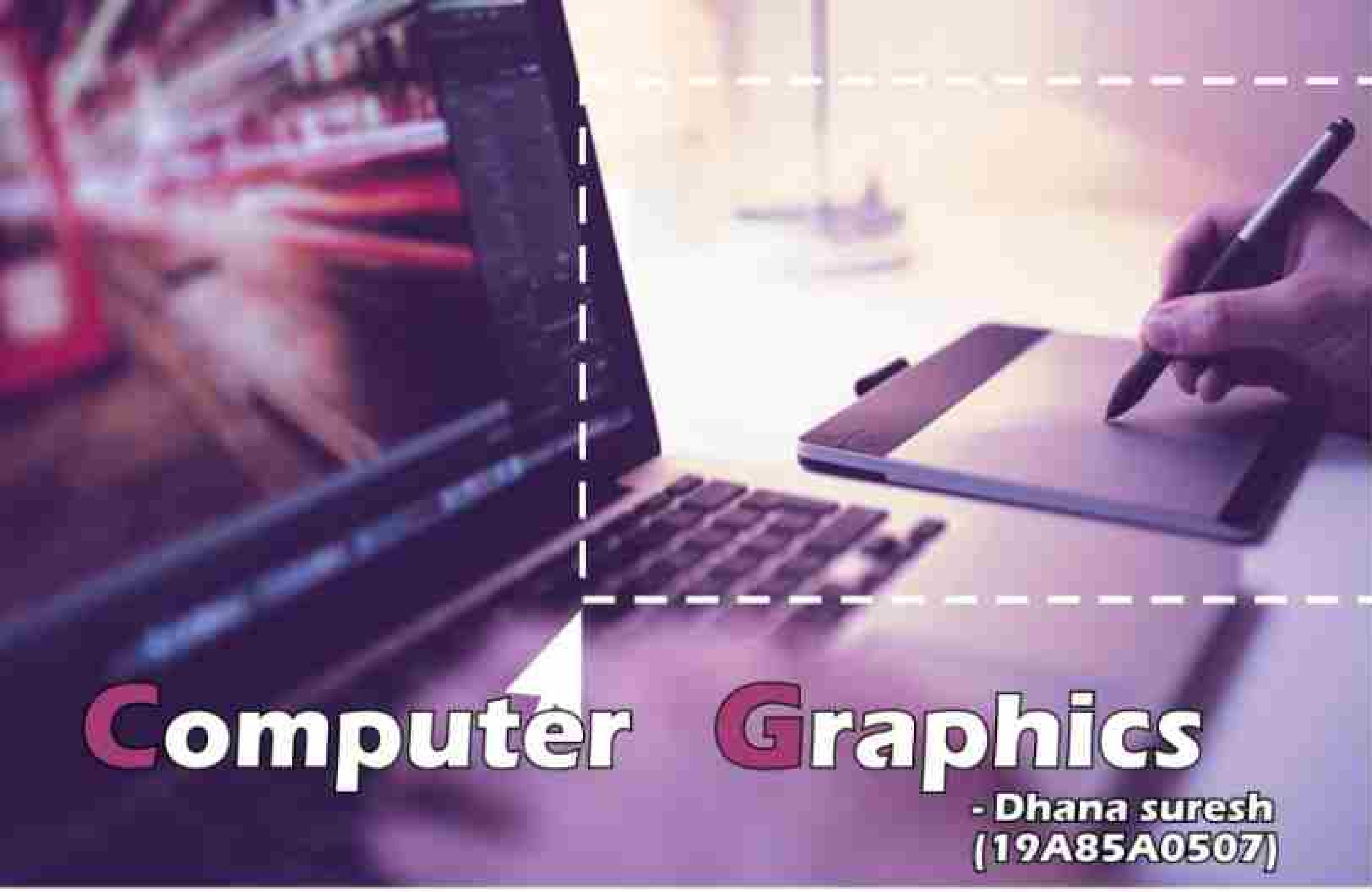
CSE MAGAZINE -VOL 10- ISSUE 01





# Computer Graphics

- Dhana suresh
(19A85A0507)

## *Introduction:-*

The creation of computer graphics has evolved dramatically as computers have become commonplace in homes around the world. Computers today are more powerful, smaller and quieter than those in the past.
The precursor sciences to the development of modern computer graphics were the advances in electrical engineering, electronics, and television that took place during the first half of the twentieth century. Screens could display art since the Lumiere brothers' use of mattes to create special effects for the earliest films dating from 1895, but such displays were limited and not interactive.
So, the first cathode ray tube, the Braun tube, was invented in 1897 – it in turn would permit the oscilloscope and the military control panel – the more direct precursors of the field, as they provided the first two-dimensional electronic displays that responded to programmatic or user input.

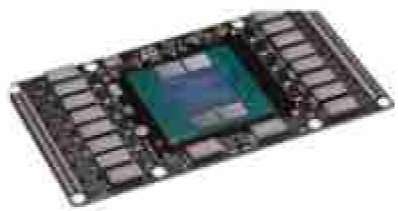
## *Graphics Tools:-*

Hear when it comes to Graphics Tools they are of two types

1) Hardware tools
2) Software tools

## Hardware tools:-

Hardware tools are very crucial for computer graphics, the main graphics hardware tool is graphics processing unit (GPU).
Graphics hardware is computer hardware that generates computer graphics and allows them to be shown on a display, usually using a graphics card (video card) in combination with a device driver to create the images on the screen.



A graphics processing unit (GPU) is a specialized electronic circuit designed to rapidly manipulate and alter memory to accelerate the creation of images in a frame buffer intended for output to a display device. GPUs are used in embedded systems, mobile phones, personal computers, workstations, and game consoles. Modern GPUs are very efficient at manipulating computer graphics and image processing. Their highly parallel structure makes them more efficient than general-purpose central processing units (CPUs) for algorithms that process large blocks of data in parallel. In a personal computer, a GPU can be present on a video card or embedded on the motherboard. In certain CPUs, they are embedded on the CPU die.

## Software tools:-

**Mainly this software tools are used to create or design graphics in computer (technically these are called as graphics modeling tools).**

### 1)Blender

**For CG artists on a budget, it doesn't get any better than Blender, the free modelling, texturing, animation and rendering app. The long-awaited version 2.8 (due for launch July 2019) provides a modern, more consistent interface, plus high-quality viewport, real-time interactive rendering, and tons of fixes and features.**
**The open-source program has been around for a long time now, and subsequently has an army of artists, teachers and enthusiasts behind its continued development. It boasts a highly impressive 3D modelling and sculpting toolset, and is considered a completely viable alternative to paid modelling programs. Blender was notorious for its non-standard way of working, but 2.8 solves a lot of these issues, and so it will feel more familiar if you're moving from an existing app.**
**Blender is a brilliant starting point to see if 3D graphics are for you – and we have a host of fantastic Blender tutorials to get started with elsewhere on the site. Despite the non-existent price tag, it's capable of producing images and animation that on a par with just about any other 3D modelling software on the market.**

## 2)Maya

Ask any 3D artist to name the best 3D modelling software, and most will choose Autodesk Maya. Largely seen as the industry standard for CG, Autodesk Maya boasts an unrivalled range of tools and features. This hugely extensible app isn't for the faint-hearted: its toolset is hugely complex and takes time to learn. However, if you're aiming to get a job in the animation or VFX industries, you'd be wise to use the same software that the likes of ILM, Pixar, DNEG and Framestore use.
Maya is great at modelling, texturing, lighting and rendering – its vast feature set includes particles, hair, solid body physics, cloth, fluid simulations and character animation. There's a chance you may never touch some of its functionality, so you need to decide if it's actually overkill for your specific needs.
This level of power also comes at a price – a subscription to Maya doesn't come cheap. But for those who have the time, skill and patience to master it, Maya has some of the best 3D tools around and is a sound investment.

## 3.ZBrush

ZBrush is a standalone sculpting and modelling app that is best suited to the creation of organic forms – although recent updates have gradually improved its hard-surface abilities. It works in a non-standard fashion, with a workflow and user interface that's initially very hard to learn, so you really need to get ZBrush and use it every day to become proficient.
However, ZBrush isn't only for sculpting and modelling: it can also be used to create UV maps and paint textures, enabling seasoned artists to craft entire figures, with clothing and props, ready for rendering. ZBrush is a popular choice among artists wanting to 3D print toys and action figures, too, with tools specifically aimed at 3D printing.

## 4) Houdini 17.5

3D modelling software is Houdini by SideFX. Widely used in the VFX industry for creating a range of amazing 3D imagery, Houdini's node-based procedural approach provides digital artists with an unprecedented level of power, flexibility and control. This nodal workflow isn't to everyone's liking, but Houdini also has more traditional tools for directly interacting with polygons on screen.
Like Maya, this level of power and non-standard workflow can be tricky to get to grips with. Fortunately, SideFX offers Houdini Apprentice, a free version of Houdini FX, which can be used by students, artists and hobbyists for personal non-commercial projects. The free version gives you access to virtually all of the features of the award-winning Houdini FX to develop your skills and for working on personal projects. The full-featured Houdini Indie also provides an affordable commercial option for small studios.

## Graphics Terminology:

1) **CAD (Computer Aided Drawing)**
2) **The Mesh**
3) **3D Rendering**
4) **Rapid Prototyping**
5) **Ambient Lighting**
6) **Beveling**
7) **Texture Mapping**
8) **3D Sculpturing**
9) **Polygonal Modeling**
10) **NURMS Modeling**

## Conclusion:-

**This is all about the computer graphics in short and about some of the tools used to create computer graphics and its terminology.**


# Array Of Things(AOT)

## Introduction:

The Array Of Things is a collaborative effort to collect the real-time data on environment,infrastructure and public use

- It is a networked sensor system.

## Why AOT is used ?

- The Array of Things (AoT) is an urban sensing network of programmable, modular nodes that will be installed around cities to collect real-time data on the city's environment, infrastructure, and activity for research and public use. AoT will essentially serve as a "fitness tracker" for the city, measuring factors that impact livability in cities such as climate, air quality and noise.

## Goals of AOT:

- AoT will provide real-time, location-based data about urban environment, infrastructure and activity to researchers and the public. This initiative has the potential to allow researchers, policymakers, developers and residents to work together and take specific actions that will make cities healthier, more efficient and more livable

The data will help cities operate more efficiently and realize cost savings by anticipating and proactively addressing challenges such as urban flooding and traffic safety Because the data will be published openly and without charge, it will also support the development of innovative applications, such as a mobile application that allows a resident to track their exposure to certain air contaminants, or to navigate through the city based on avoiding urban heat islands, poor air quality, or excessive noise and congestion.

## Tecnology used in AOT:

- Array of Things also serves as the flagship deployment of an innovative new type of cyberinfrastructure -- a distributed, programmable system of nodes that can be used to answer critical research questions across different settings and fields of study.
- AoT is based upon Waggle technology, an open platform for edge computing and intelligent, wireless sensors developed at Argonne National Laboratory. In addition to AoT and other urban research initiatives, Waggle software and hardware supports environmental and atmospheric science in a variety of environments.

## Data collected by AOT:

• The nodes will initially measure temperature, barometric pressure, light, vibration, carbon monoxide, nitrogen dioxide, sulfur dioxide, ozone, ambient sound intensity, and pedestrian and vehicle traffic. Continued research and development is using machine learning to create sensors to monitor other urban factors of interest such as flooding and standing water.

• It has a privacy protection is built into the design of the sensors and into the operating policies.

## Data availability on device: •

Data collected by AoT is open, free, and available to the public. The nodes transmit data to a secure central database server at Argonne National Laboratory(in Chicago). Data is then published openly to allow individuals, organizations, researchers, engineers and scientists to study urban environments, develop new data analysis tools and applications, and inform urban planning.

## Data security in AOT:

• Policy and operational activities are guided by the Array of Things,EOC committee that's under the government.

• Any changes to privacy, such as additional image processing algorithms or sensors that could potentially have privacy implications, require the approval of the government and then the AoT EOC, as outlined in the operation of Array of Things is governed by our privacy policies.

• The EOC draws upon external technical security and privacy expertise through the Technical Security and Privacy Group and Operating as an external, independent review team, the committee will also be consulted whenever there is a request for a new kind of data to be collected.

### Applicatons of AOT:

• Sensors monitoring air quality, sound and vibration (to detect heavy vehicle traffic), and temperature can be used to suggest the healthiest and unhealthiest walking times and routes through the city, or to study the relationship between diseases and the urban environment.

• Real-time detection of urban flooding can improve city services and infrastructure to prevent property damage and illness.

• Measurements of micro-climate in different areas of the city, so that residents can get up-to-date, high-resolution "block-by-block" weather and climate information.

• Observe which areas of the city are heavily populated by pedestrians at different times of day to suggest safe and efficient routes for walking late at night or for timing traffic lights during peak traffic hours to improve pedestrian safety and reduce congestion-related pollution.

## How nodes are installed:

• This nodes are installed to an electric poles,traffic signal poles,and industry etc.... pubilc places

• This nodes stores the data by using the sensors devices

• These sensors do not have the capability to measure or identify individuals.

• Microphones and cameras in public spaces do not collect sensitive personally identifiable information (PII). Microphone and camera images will be processed in near-real-time within the installed equipment, not transmitted or stored, with the exception of less than 1% of images at random times, saved for the purposes of image processing software calibration. Although these images will not contain PII, they will be controlled and protected with the same measures typically afforded PII.

## Sensors used:

• Honey well • SPEC Sensors 3SP-CO-1000 • SPEC Sensors IAQ-100 ect....

# Ethical Hacking

CH.H.V.Siddartha Varma
(18A81A05I5)

## INTRODUCTION:

Today's Technology is playing a crucial role in the Whole world. So every person has minimum Knowledge of technology. The thief in the present society doesn't require any keys but he should require a small information that related to every individual, so that he can do anything by showing your information.
"INFORMATION IS EVERYTHING"

## 1.Phishing:

Phishing is a cyber attack that uses disguised email as a weapon. The goal is to trick the email recipient into believing that the message is something they want or need a request from their bank, for instance, or a note from someone in their company and to click a link or download an attachment.

What really distinguishes phishing is the form the message takes: the attackers masquerade as a trusted entity of some kind, often a real or plausibly real person, or a company the victim might do business with. It's one of the oldest types of cyberattacks, dating back to the 1990s, and it's still one of the most widespread and pernicious, with phishing messages and techniques becoming increasingly sophisticated.

## PLATFORMS:

There are many platforms to do hacking. In that mainly

Kali Linux: PARROT OS:

## HACKERS:

There are basically 3 types of hackers, they are

1. White Hat's :
Who use their skills for Nation.

2. Black Hat's :
Who use their skills for illegal purpose.

3. Grey Hat's :
Both Black and White Hat's.

## 2.Brute-Force attack:

In cryptography, a brute force attacker consists of an attacker submitting many passwords or passphrases with the hope of eventually guessing correctly. The attacker systematically checks all possible passwords and passphrases until the correct one is found.

Alternatively, the attacker can attempt to guess the key which is typically created from the password using a key derivation function. This is known as an exhaustive key search.

A brute-force attack is a cryptanalytic attack that can, in theory, be used to attempt to decrypt any encrypted data (except for data encrypted in an information-theoretically secure manner). Such an attack might be used when it is not possible to take advantage of other weaknesses in an encryption system (if any exist) that would make the task easier.

When password-guessing, this method is very fast when used to check all short passwords, but for longer password

s other methods such as the dictionary attack are used because a brute-force search takes too long. Longer passwords, passphrases and keys have more possible values, making them exponentially more difficult to crack than shorter ones.

Brute-force attacks can be made less effective by obfuscating the data to be encoded making it more difficult for an attacker to recognize when the code has been cracked or by making the attacker do more work to test each guess. One of the measures of the strength of an encryption system is how long it would theoretically take an attacker to mount a successful brute-force attack against it.

Brute-force attacks are an application of brute-force search, the general problem-solving technique of enumerating all candidates and checking each one.

## Hacking Techniques:

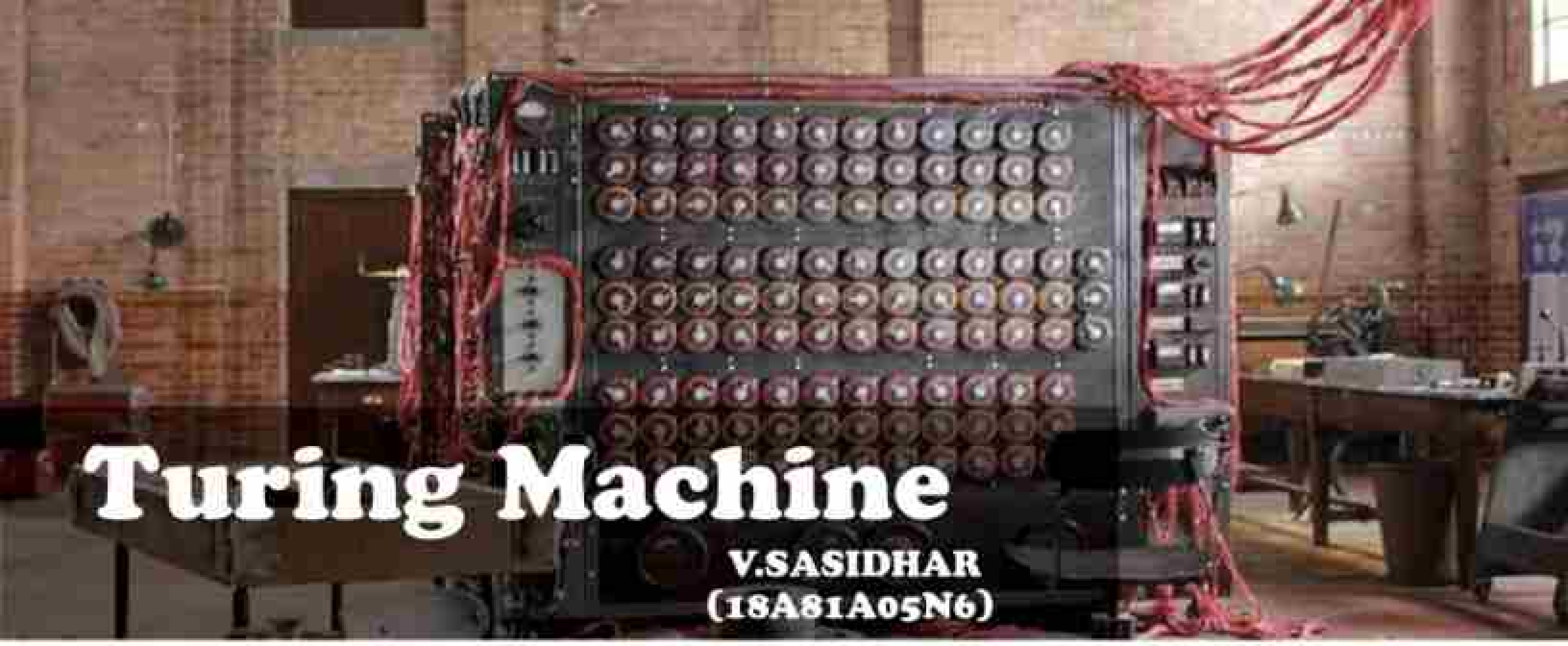
There are many techniques in hacking in that mainly

1. phishing
2. Brute-Force attack:

## 1. phishing



## 2. Brute-Force attack:



# Turing Machine

V.SASIDHAR
(18A81A05N6)

## Formal definition of Turing machine :

We formalize Turing's description as follows: A Turing machine consists of a finite program, called the finite control, capable of manipulating a linear list of cells, called the tape, using one access pointer, called the head. We refer to the two directions on the tape as `right' and `left.' The finite control can be in any one of a finite set of states Q , and each tape cell can contain a 0, a 1, or a `blank' B . Time is discrete and the time instants are ordered 0,1,2,..., with 0 the time at which the machine starts its computation. At any time, the head is positioned over a particular cell, which it is said to `scan.' At time 0 the head is situated on a distinguished cell on the tape called the `start cell,' and the finite control is in a distinguished state q0 . At time 0 all cells contain B's, except for a contiguous finite sequence of cells, extending from the start cell to the right, which contain 0's and 1's. This binary sequence is called the `input.' The device can perform the following basic operations:

1. it can write an element from A={0,1,B} in the cell it scans; and
2. it can shift the head one cell left or right.

A Turing machine refers to a hypothetical machine proposed by Alan M. Turing (1912--1954) in 1936 whose computations are intended to give an operational and formal definition of the intuitive notion of computability in the discrete domain. It is a digital device and sufficiently simple to be amenable to theoretical analysis and sufficiently powerful to embrace everything in the discrete domain that is intuitively computable. As if that were not enough, in the theory of computation many major complexity classes can be easily characterized by an appropriately restricted Turing machine; notably the important classes P and NP and consequently the major question whether P equals NP.

When the device is active it executes these operations at the rate of one operation per time unit (a `step'). At the conclusion of each step, the finite control takes on a state from Q . The device is constructed so that it behaves according to a finite list of rules. These rules determine, from the current state of the finite control and the symbol contained in the cell under scan, the operation to be performed next and the state to enter at the end of the next operation execution.

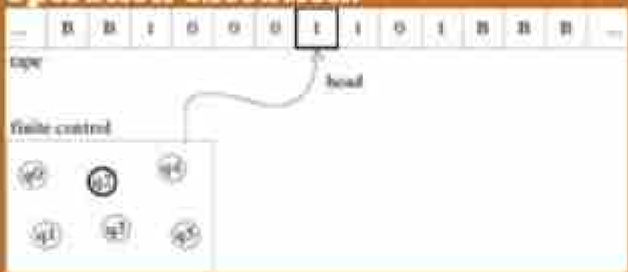



Figure : Turing machine

The rules have format (p,s,a,q) : p is the current state of the finite control; s is the symbol under scan; a is the next operation to be executed of type (1) or (2) designated in the obvious sense by an element from S={0,1,B,L,R} ; and q is the state of the finite control to be entered at the end of this step.
For now, we assume that there are no two distinct quadruples that have their first two elements identical, the device is deterministic. Not every possible combination of the first two elements has to be in the set; in this way we permit the device to perform `no' operation. In this case we say that the device halts. Hence, we can define a Turing machine by a mapping from a finite subset of $Q \times A$ into $S \times Q$ . Given a Turing machine and an input, the Turing machine carries out a uniquely determined succession of operations, which may or may not terminate in a finite number of steps. Strings and natural numbers are occasionally identified according to the pairing (E,0),(0,1),(1,2),(00,3),(01,4),(10,5),(11,6),...,

## Computable functions

We can associate a partial function with each Turing machine in the following way: The input to the Turing machine is presented as an n-tuple (x1,...,xn) consisting of self-delimiting versions of the xi's. The integer represented by the maximal binary string (bordered by blanks) of which some bit is scanned, or 0 if a blank is scanned, by the time the machine halts, is called the `output' of the computation. Under this convention for inputs and outputs, each Turing machine defines a partial function from n-tuples of integers onto the integers, $n \geq 1$ . We call such a function partial computable. If the Turing machine halts for all inputs, then the function computed is defined for all arguments and we call it total computable. (Instead of `computable' the more ambiguous `recursive' has also been used.) We call a function with range {0,1} a `predicate', with the interpretation that the predicate of an n-tuple of values is `true' if the corresponding function assumes value 1 for that n-tuple of values for its arguments and is `false' or `undefined' otherwise. Hence, we can talk about `partial (total) computable predicates'.

## Examples of computable functions

Consider x as a binary string. It is easy to see that the functions |x| , f(x)=x, g(x,y)=x , and h(x,y)=y are partial computable. Functions g and h are not total since the value for input 1111 is not defined. The function g(x,y) defined as 1 if x=y and as 0 if $x \neq y$ is a computable predicate. Consider x as an integer. The following functions are basic n-place total computable functions: the `successor' function γ(1)(x)=x+1 , the `zero' function g(n)(x1,...,xn)=0 , and the `projection' function π(n)m(x1,...,xn)=xm ($1 \leq m \leq n$). The function (x,y)=xy is a total computable one-to-one mapping from pairs of natural numbers into the natural numbers. We can easily extend this scheme to obtain a total computable one-to-one mapping from k-tuples of integers into the integers, for each fixed k . Define (n1,n2,...,nk) =(n1,(n2,...,nk)) . Another total recursive one-to-one mapping from k-tuples of integers into the integers is (n1,n2,...,nk)=n1...nk−1nk .

Ref. No. SVEC/CSE/Reports/2019/01 Date:01/06/2019

# CSE Progress Report from 1st June 2019 to August 2019

1) **Details of faculty attended FDPs, Workshops, Seminars, Conferences etc., outside the college as well as in the college:**

**(a) FDPs / Workshops Attended by Faculty:12**

| S.No | Name and Designation of the Faculty | Name of Workshop/Seminar/ FDP/SDP Attended | Location | No. Of Days | From Date | To Date |
|---|---|---|---|---|---|---|
| 1. | D Anjani Suputri Devi | Digital image processing for medical images | SRKR Institute, Bhimavaram | 10 | 05-08-2019 | 17-08-2019 |
| 2. | A Lakshmi Lavanya | Blockchain & Bitcoin applications | Adikavi Nannayya University, Rajhamundry | 02 | 05-07-2019 | 06-07-2019 |
| 3. | D Anjani Suputri Devi | Research Methodology | SVEC | 01 | 01-07-2019 | 01-07-2019 |
| 4. | B Madhava Rao | Research Methodology | SVEC | 01 | 01-07-2019 | 01-07-2019 |
| 5. | D Sasi rekha | Research Methodology | SVEC | 01 | 01-07-2019 | 01-07-2019 |
| 6. | Dr. D Jaya Kumari | Assessment &Evaluation under Outcome Based Education | SVEC | 05 | 10-06-2019 | 14-06-2019 |
| 7. | Y Ravi Raju | Assessment &Evaluation under Outcome Based Education | SVEC | 05 | 10-06-2019 | 14-06-2019 |
| 8. | M Nageswara Rao | Assessment &Evaluation under Outcome Based Education | SVEC | 05 | 10-06-2019 | 14-06-2019 |
| 9. | M R Raja Ramesh | Assessment &Evaluation under Outcome Based Education | SVEC | 05 | 10-06-2019 | 14-06-2019 |
| 10. | N Praveen Kumar | Assessment &Evaluation under Outcome Based Education | SVEC | 05 | 10-06-2019 | 14-06-2019 |
| 11. | G Loshma. | Deep Learning | NIT, Andhra Pradesh | 06 | 03-06-2019 | 08-06-2019 |
| 12. | Dr. K Shirin Bhanu | Deep Learning | NIT, Andhra Pradesh | 06 | 03-06-2019 | 08-06-2019 |

**2 (a) Workshops/FDPs/Seminars etc. conducted by the Department to the students: 01**

| S.No | Date | Event Name | Name of the Eminent Guest | Audience | No of Students participated |
|---|---|---|---|---|---|
| 1 | 22.07.2019 to 27.07.2019 | Web Development with Python | 1. B Ayyappa<br>2. B Umamahesh from APSSDC | III CSE | 86 |

2(b) **Training Programmes conducted for students: 04**

| S.No | Name of the Event | Speaker Institute | Date | Audience | Organized By |
|---|---|---|---|---|---|
| 1 | **CRT Training (Mphasis Company Specific)** | Talentio | 27.07.2019 – 03.08.2019 | IV CSE Students | College |
| 2 | **CRT Training (Company Specific)** | Talentio | 20.07.2019 – 24.07.2019 | IV CSE Students | College |
| 3 | **CRT Training (Company Specific)** | Talentio | 08.07.2019 – 19.07.2019 | IV CSE Students(169) | College |
| 4 | **CRT Training** | Talentio | 03.06.2019 – 13.06.2019 | IV CSE Students(69) | College |

**2(c) Guest Lectures conducted for students: 02**

| S.No | Name of the Event | Speaker institute | Date | Audience | Organized By |
|---|---|---|---|---|---|
| 1 | **Guest Lecture on Global Infrastructure services** | AVP - Hosting | 27.08.2019(FN) | IV CSE Students | College |
| 2 | **Guest Lecture on Global Infrastructure services** | AVP - Hosting | 27.08.2019(AN) | III CSE Students | College |

4) **Details of publications of faculty: 02**

| S.No. | Name of the Faculty | Designation | Publication Title | Publication Details | INDEXING SCI/SCOPUS/UGC LISTED/OTHERS |
|---|---|---|---|---|---|
| 1 | Dr V Venkateswara Rao | Professor | Prediction and Analysis of sentiments on Twitter Data using Hybrid Naïve Bayes Approach | International journal of Innovative technology and exploring engineering, Volume 8, Issue 8, June 2019 | SCOPUS |
| 2 | Dr.D.Jaya Kumari | Professor | Network Based Adaptation of Block Chain Technology | International Journal of Innovative Technology and Exploring Engineering (IJITEE) ISSN: 2278-3075, Volume-8 Issue-9, July 2019 | SCOPUS |

5) B.Tech III SEM (Autonomous) Class work started on 10.06.2019.
6) B.Tech III-I & IV-I (R16) Class work started on 10.06.2019.
7) JNTUK FFC visit on 15.06.19.
8) Orientation Programme for III Sem Students conducted by Dean(S) & HOD on 08.07.2019.

9) **Student Achievements:**

(a) On the occasion of SBI foundation day, SBI honored the college toppers on 01.07.2019.

| S.No. | HT Number | Name | Branch | CGPA |
|---|---|---|---|---|
| 1 | 18A81A05J8 | KALLA GAYATRI | CSE | 9.92 |

## b) <u>Workshops Attended by Students(Inter College Level) :03</u>

| S.No | Regd.No. | Name Of The Student | Branch | Name Of The Event | Month-Date |
|---|---|---|---|---|---|
| 1 | 18A81A0535 | N.J.N.V.S.K.S.V.Challa Rao | CSE | AI workshop conducted by ETHICAL EDUFABRICA PVT.LTD at NIT –AP, Tadepalligudem | 17-08-2019 - 18-08-2019 |
| 2 | 18A81A05K4 | K.Uma Sasank | | | |
| 3 | 18A81A0591 | Swathi Sri Naryana | | | |

## C) <u>Workshops Attended by Students(Intra College Level) :51</u>

| S.No | Regd.No. | NAME OF THE STUDENT | Branch | NAME OF THE EVENT | Month-Date |
|---|---|---|---|---|---|
| 1. | 18A85A05I7 | D.Sahithi | CSE | 3 Days Entrepreneurship Awareness Camp- Supported by EDII & NSTEDB, DST, New Delhi held at Sri Vasavi Engineering College. | 29-08-2019 - 31-08-2019 |
| 2. | 18A85A05N3 | U.Gamya | | | |
| 3. 1 | 18A85A05J6 | I.Lakshmi prasanna | | | |
| 4. | 18A85A0578 | K.Bhavya | | | |
| 5. | 18A85A0507 | Satish | | | |
| 6. | 18A85A05K0 | K.Surekha | | | |
| 7. | 18A85A05I6 | CH.Jai Krishna | | | |

| | | | | | |
|---|---|---|---|---|---|
| 8. | 18A85A05J4 | Manikanta | | | |
| 9. | 18A81A05I3 | M. Ramya Sri | | | |
| 10. | 17A81A05I2 | V.Subra Manyeswari | | | |
| 11. | 17A81A05J7 | Dharani | | | |
| 12. G | 18A85A05G4 | G.Purna Sri | | | |
| 13. | 18A85A05A4 | S.Ramya | | | |
| 14. | 18A85A05K2 | K.Eswari | | | |
| 15. | 19A85A0505 | S.Krishna Prasad | | | |
| 16. | 19A85A0503 | K.Avinash | | | |
| 17. | 17A81A0598 | Akanksha | | | |
| 18. | 17A81A05J6 | M.Geetha | | | |
| 19. | 17A81A0595 | P.Bhavani | | | |
| 20. | 17A81A05I2 | G.Sahadev | | | |
| 21. | 17A81A05I5 | J.Pragathi | | | |
| 22. | 17A81A05G7 | Y.S.G.S.S.Bhavani | CSE | 3 Days Entrepreneurship Awareness Camp-Supported by EDII & NSTEDB, DST, New Delhi held at Sri Vasavi Engineering College. | 29-08-2019 - 31-08-2019 |
| 23. | 17A81A05L4 | V.Prathyusha | | | |
| 24. | 17A81A05E8 | P.Dhana Lakshmi | | | |
| 25. | 17A81A05G3 | V.Subhashini | | | |
| 26. | 17A81A05F6 | B.Uttej | | | |
| 27. | 17A81A05G4 | Y.Purna Sainath | | | |
| 28. | 17A81A05L5 | V.Badrinath | | | |
| 29. | 17A81A05J5 | M.Gopi | | | |
| 30. | 17A81A05K8 | S.Rakesh | | | |
| 31. | 17A81A05H0 | B.Narayana | | | |
| 32. | 17A81A05H6 | CH.Sriram | | | |

| | | | | | |
|---|---|---|---|---|---|
| 33. | 17A81A05F1 | P.Sahithi | | | |
| 34. | 17A81A05F4 | R.Sai Kiran | | | |
| 35. | 17A81A05E0 | M.Heamanth | | | |
| 36. | 16A81A05H0 | P.Anirudh | | | |
| 37. | 17A81A05E6 | N.Rakesh | | | |
| 38. | 17A81A05D2 | G.Prasad Babu | | | |
| 39. | 17A81A05F2 | S.Nagesh | | | |
| 40. | 17A81A05D2 | G.Srija | CSE | 3 Days Entrepreneurship Awareness Camp-Supported by EDII & NSTEDB, DST, New Delhi held at Sri Vasavi Engineering College. | 29-08-2019 - 31-08-2019 |
| 41. | 17A81A05F9 | M.Vardhan | | | |
| 42. | 17A81A05G2 | V.Prasanna Lakshmi | | | |
| 43. | 17A81A05E2 | M.Yaswanth Kumar | | | |
| 44. | 17A81A05K6 | S.Mahendra | | | |
| 45. | 18A81A05A8 | Nasrin | | | |
| 46. | 17A81A05E7 | P.K.V.S.S.Viswanath | | | |
| 47. | 19A85A0521 | S.Naga Phanindra | | | |
| 48. | | K.Bhavani | | | |
| 49. | 17A81A05G5 | Y.Mounika | | | |
| 50. | 18A81A05I1 | Satish | | | |
| 51. | 17A81A05A4 | S Anusha | CSE | 3 Days Entrepreneurship Awareness Camp-Supported by EDII & NSTEDB, DST, New Delhi held at Sri Vasavi Engineering College. | 29-08-2019 - 31-08-2019 |

### C) Co-Curricular Activities:

| S.NO. | Roll Number | Name of the Student | Name of the event | Venue | Duration | Team Position |
|---|---|---|---|---|---|---|
| 1 | 18A81A0538 | N.Seshasai | Shuttle Badminton | DRMC indoor stadium, Vijayawada | 21-8-2019 to 23-8-2019 | 1st place |
| 2 | 17A81A05L5 | V.Badrinath | | | | 4th place |

10)Sahaya: As part of Sahaya the following list of events are conducted

| S.No | Date | Service Activity Details | Venue |
|---|---|---|---|
| 1. | 24/07/2019 | The team of SAHAYA donated Rs. 8200(Eight thousands two hundred ) to Mr. Yeswanth suffered with Brain hemorrhage due to sudden attack | Sri Vasavi Engg. College, TPG |

# Theory Of Multiverse

M. Sri Charan
(18A81A05F4)

For Every realtiy there is alternate suitable reality may exits with in the existing universe
So every possible out come of time has another possibility which results another reality co exists with in the same incident so whenever we want to do something we can assume that somewhere in alternate reality we haven't done that thing
So how this multiverse happen . In space there will be warmholes and blackholes which are created by singularity so blackhole will have the large gravitonal pull which allows that to bend space so where both of it are realtive the time around the backhole will get fluctuated .so those fluctuations effect the flow of time in the space effecting the cosmic entities of time altering events effecting alternate possibility called multiverse

3d view of a 4th dimentional object

**Multiverses are predictions based on theories. Focusing on each theory's assumptions is key to evaluating a proposed multiverse. Although accepted theories of particle physics and cosmology contain non-intuitive features, multiverse theories entertain a host of "strange" assumptions classified as metaphysical (outside objective experience, concerned with fundamental nature of reality, ideas that cannot be proven right or wrong) topics such as: infinity, duplicate yous, hypothetical fields, more than three space dimensions, Hilbert space, advanced civilizations, and reality established by mathematical relationships. It is easy to confuse multiverse proposals because many divergent models exist. This overview defines the characteristics of eleven popular multiverse proposals. The characteristics compared are: initial conditions, values of constants, laws of nature, number of space dimensions, number of universes, and fine tuning explanations. Future scientific experiments may validate selected assumptions; but until they do, proposals by philosophers may be as valid as theoretical scientific theories. Keywords: multiverse, metaphysics, philosophy, inflation, string theory, cyclic universe, Loop Quantum Gravity, simulated universe, ultimate universe, fine tuning, infinity**

# Proposals on multiverse

## Proposal one :

The Finite Quilted multiverse exists within a finite expanse of space, simply more of what we observe, each universe like a patch replicated on a quilt. There may exist hundreds or thousands of Hubble volumes. These parallel universes embody similar initial conditions, the same constants of nature, the same laws of physics, and three space dimensions. The Finite Quilted multiverse adds no insight to fine tuning. In the 1920's, Alexander Friedmann showed that a variety of expanding, contracting, or oscillating universes were compatible with Einstein's General Theory of Relativity. At about the same time, George Lemaitre predicted that our universe began with a big bang – everything we observe originated in a hot, dense, and rapidly expanding space.

## Proposal two:

The Infinite Quilted multiverse is similar to the Finite Quilted multiverse except for different initial conditions and the number of parallel universes – instead of thousands of versions, an infinite number of universes populate space and therefore fine tuning can be explained. Is space finite or infinite? Most multiverses assume an infinite space where reality takes on surprising new meanings – for example, the prediction of duplicate yous (or infinite number of yous) living in an identical world somewhere in another universe. Einstein's equations for general relativity allow for either a finite or an infinite spatial extent. All cosmological models involve an expanding universe, but they differ in the overall shape of space and whether space is finite or infinite [Greene 2011, 23].

## Proposal three:

In the Inflationary multiverse, the eternal (chaotic) inflation theory produces different constants of nature in separate universes. Sometimes referred to as a bubble, each universe is separated by expanding space. Inflation produces an infinite number of universes, and thus provides an explanation for the presence of fine tuning. Inflationary cosmology is not one unique theory, rather it provides a framework containing many versions differing in details, such as number of inflaton fields (spelled differently than inflation) and their potential-energy curves. However, in general, the basic inflation concept

## Mathematical Universes

Scientists have debated whether mathematics is simply a useful tool for describing the universe, or whether math itself is the fundamental reality, and our observations of the universe are just imperfect perceptions of its true mathematical nature. If the latter is the case, then perhaps the particular mathematical structure that makes up our universe isn't the only option, and in fact all possible mathematical structures exist as their own separate universes.

"A mathematical structure is something that you can describe in a way that's completely independent of human baggage," said Max Tegmark of MIT, who proposed this brain-twistin gidea. "I really believe that there is this universe out there that can exist independently of me that would continue to exist even if there were no humans."

## Bubble Universes

In addition to the multiple universes created by infinitely extending space-time, other universes could arise from a theory called "eternal inflation." Inflation is the notion that the universe expanded rapidly after the Big Bang, in effect inflating like a balloon. Eternal inflation, first proposed by Tufts University cosmologist Alexander Vilenkin, suggests that some pockets of space stop inflating, while other regions continue to inflate, thus giving rise to many isolated "bubble universes."

## Parallel Universes:

Another idea that arises from string theory is the notion of "braneworlds" — parallel universes that hover just out of reach of our own, proposed by Princeton University's Paul Steinhardt and Neil Turok of the Perimeter Institute for Theoretical Physics in Ontario, Canada. The idea comes from the possibility of many more dimensions to our world than the three of space and one of time that we know. In addition to our own three-dimensional "brane" of space, other three-dimensional branes may float in a higher-dimensional space.

# The Treasure House Within You

V.SURYA KAMAL
(18A81A05N5)

Infinite Riches are all around you if you will open your mental eyes and behold the treasure house of infinity within you. There is a gold mine within you from which you can extract everything you need to live life gloriously, joyously and abundantly. Many are sound asleep because they do not know about this gold mine of Infinite Intelligence and boundless love within themselves. Whatever you want, you can draw forth. A magnetized piece of steel will lift about twelve times its own weight, and if you demagnetize this same piece of steel, it will not even lift a feather. Similarly, there are two types of men. There is the magnetized man who is full of confidence and faith. He knows that he is born to win and to succeed. Then, there is the type of man who is demagnetized. He is full of fears and doubts. Opportunities come, and he says, "I might fail; I might lose money; people will laugh at me." This type of man will not get very far in life because, if he is afraid to go forward, he will simply stay where he is. Become a magnetized man and discover the master secret of the ages

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## The Master Secret Of The Ages

What, in your opinion, is the master secret of the ages? The secret of atomic energy? Thermonuclear energy? The neutron bomb? Interplanetary travel? No, not any of these. Then, what is this master secret? Where can one find it, and how can it be contacted and brought into action? The answer is extraordinarily simple. This secret is the marvelous, miracle-working power found in your own subconscious mind, the last place that most people would seek it.

## The Marvelous Power Of Your Subconscious Mind

You can bring into your life more power, more wealth, more health, more happiness and more joy by learning to contact and release the hidden power of your subconscious mind.

You need not acquire this power; you already possess it. But, you want to learn how to use it; you want to understand it so that you can apply it in all departments of your life.

As you follow the simple techniques and processes set forth in this book, you can gain the necessary knowledge and understanding. You can be inspired by a new light, and you can generate a new force enabling you to realize your hopes and make all your dreams come true. Decide now to make your life grander, greater, richer and nobler than ever before.

Within your subconscious depths lies infinite wisdom, infinite power and infinite supply of all that is necessary, which is waiting for development and expression. Begin now to recognize these potentialities of your deepMore Free Books Law of Attraction Havener mind and they will take form in the world without.

The Infinite Intelligence within your subconscious mind can reveal to you everything you need to know at every moment of time and point of space provided you are open minded and receptive. You can receive new thoughts and ideas enabling you to bring forth new inventions, make new discoveries or write books and plays. Moreover, the Infinite Intelligence in your subconscious can impart to you wonderful kinds of knowledge of an original nature. It can reveal to you and open the way for perfect expression and true place in your life.

Through the wisdom of your subconscious mind you can attract the ideal companion, as well as the right business associate or partner. It can find the right buyer for your home and provide you with all the money you need and the financial freedom to be, to do, and to go as your heart desires.

It is your right to discover this inner world of thought, feeling and power, of light, love and beauty. Though invisible, its forces are mighty. Within your subconscious mind you will find the solution for every problem and the cause for every effect. You can draw out the hidden powers; you come into actual possession of the power and wisdom necessary to move forward in abundance, security, joy and dominion.

I have seen the power of the subconscious lift people up out of crippled states, making them whole, vital and strong once more, free to go out into the world to experience happiness, health and joyous expression.

There is a miraculous healing power in your subconscious that can heal the troubled mind and the broken heart. It can open the prison door of the mind and liberate you. It can free you from all kinds of material and physical bondage.

## Necessity Of A Working Basis

Substantial progress in any field of endeavor is impossible in the absence of a working basis, which is universal in its application. You can become skilled in the operation of your subconscious mind. You can practice its powers with a certainty of results in exact proportion to your knowledge of its principles and to your application of them for definite, specific purposes and goals you wish to achieve.

Being a former chemist, I would like to point out that if you combine hydrogen and oxygen in the proportions of two atoms of the former to one of the latter, water will be the result. You are very familiar with the fact that one atom of oxygen and one atom of carbon will produce carbon monoxide, a poisonous gas. But, if you add another atom of oxygen, you will get carbon dioxide, a harmless gas, and so on throughout the vast realm of chemical compounds.

You must not think that the principles of chemistry, physics and mathematics differ from the principles of your subconscious mind. Let us consider a generally accepted principle: "Water seeks its own level." This is a universal principle, which is applicable to water everywhere.

Consider another principle: "Matter expands when heated." This is true everywhere, at any time, and under all circumstances. You can heat a piece of steel and it will expand regardless whether the steel is found in China, England or India. It is a universal truth that matter expands when healed. It is also a universal truth that whatever you impress on your subconscious mind is expressed on the screen of space as condition, experience and event.

Your prayer is answered because your subconscious mind is principle, and by principle I mean the way a thing works. For example, the principle of electricity is that it works from a higher to a lower potential. You do not change the principle of electricity when you use it, but by cooperating with nature, you can bring forth marvelous inventions and discoveries which bless humanity in countless ways.

Your subconscious mind is principle and works according to the law of belief. You must know what belief is, why it works and how it works. Your Bible says in a simple, clear and beautiful way: Whosoever shall say unto this mountain, be thou removed, and be thou cast into the sea; and shall not doubt in his heart, but shall believe that those things which he saith shall come to pass; he shall have whatsoever he saith. Mark 11:23.

The law of your mind is the law of belief. This means to believe in the way your mind works, to believe in belief itself. The belief of your mind is the thought of your mind – that is simple – just that and nothing else.

All your experiences, events, conditions and acts are the reactions of your subconscious mind to your thoughts. Remember, it is not the thing believed in, but the belief in your own mind, which brings about the result. Cease believing in the false beliefs, opinions, superstitions and fears of mankind. Begin to believe in the eternal verities and truths of life, which never change. Then, you will move onward, upward and Godward.

Whoever reads this book and applies the principles of the subconscious mind herein set forth, will be able to pray scientifically and effectively for himself and for others. Your prayer is answered according to the universal law of action and reaction. Thought is incipient action. The reaction is the response from your subconscious mind, which corresponds with the nature of your thought. Busy your mind with the concepts of harmony, health, peace and good will and wonders will happen in your life.

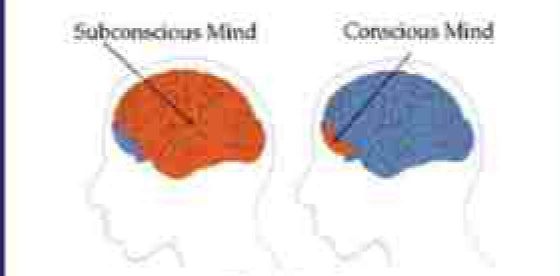
## The Duality Of Mind

You have only one mind, but your mind possesses two distinctive characteristics. The line of demarcation between the two is well known to all thinking men and women today. The two functions of your mind are essentially unlike. Each is endowed with separate and distinct attributes and powers. The nomenclature generally used to distinguish the two functions of your mind is as follows: The objective and subjective mind, the conscious and subconscious mind, the waking and sleeping mind, the surface self and the deep self, the voluntary mind and the involuntary mind, the male and the female and many other terms. You will find the terms "conscious" and "subconscious" used to represent the dual nature of your mind throughout this book.

## The Conscious & Subconscious Minds

An excellent way to get acquainted with the two functions of your mind is to look upon your own mind as a garden. You are a gardener, and you are planting seeds (thoughts) in your subconscious mind all day long, based on your habitual thinking. As you sow in your subconscious mind, so shall you reap in your body and environment.

Begin now to sow thoughts of peace, happiness, right action, good will and prosperity. Think quietly and with interest on these qualities and accept them fully in your conscious reasoning mind. Continue to plant these wonderful seeds (thoughts) in the garden of your mind and you will reap a glorious harvest. Your subconscious mind may be likened to soil, which will grow all kinds of seeds, good or bad. Do men gather grapes or thorns, figs or thistles? Every thought is, therefore, a cause, and every condition is an effect. For this reason, it is essential that you take charge of your thoughts so as to bring forth only desirable conditions.



The Mind

When your mind thinks correctly, when you understand the truth, when the thoughts deposited in your subconscious mind are constructive. Harmonious and peaceful the magic working power of your subconscious will respond and bring about harmonious conditions, agreeable surroundings and the best of everything. When you begin to control your thought processes, you can apply the powers of your subconscious to any problem or difficulty. In other words, you will actually be consciously cooperating with the infinite power and omnipotent law, which governs all things.

Look around you wherever you live and you will notice that the vast majority of mankind lives in the world without and the more enlightened men are intensely interested in the world within. Remember, it is the world within, namely your thoughts, feelings and imagery that makes your world without. It is, therefore, the only creative power and everything that you find in your world of expression has been created by you in the inner world of your mind, consciously or unconsciously.

## Brief Summary Of Ideas Worth Remembering

1. The treasure house is within you. Look within for the answer to your heart's desire.
2. The great secret possessed by the great men of all ages was their ability to contact and release the powers of their subconscious mind. You can do the same.
3. Your subconscious has the answer to all problems. If you suggest to your subconscious prior to sleep, "I want to get up at 6 a.m.," it will awaken you at that exact time.
4. Your subconscious mind is the builder of your body and can heal you. Lull yourself to sleep every night with the idea of perfect health, and your subconscious, being your faithful servant, will obey you.
5. Every thought is a cause and every condition is an effect.
6. If you want to write a book, write a wonderful play, give a better talk to your audience, then convey the idea lovingly and feelingly to your subconscious mind, and it will respond accordingly.
7. You are like a captain navigating a ship. He must give the right orders, and likewise, you must give the right orders (thoughts and images) to your subconscious mind, which controls and governs all your experiences.
8. Never use the terms, "I can't afford it" or "I can't do this." Your subconscious mind takes you at your word and sees to it that you do not have the money or the ability to do what you want to do. Affirm, "I can do all things through the power of my subconscious mind."
9. The law of life is the law of belief. A belief is a thought in your mind. Do not believe in things to harm or hurt you. Believe in the power of your subconscious to heal, inspire, strengthen and prosper you. According to your belief is it done unto you.
10. Change your thoughts and you change your destiny

# Snippet

1. **Draw one line on this equation to make it correct**

   **5+5+5+5=555**

2. **A secret can be told only to 2 persons in 5 minutes. The same person tells to two more persons and so on.**
   **How long will take to sprea a secret to 768 persons?**

3. **A clock is started at noon. By 10 minutes past 5, the hour hand has turned through:**

   **A.145 deg** **B.150 deg**
   **C.155 deg** **D.160 deg**

4. **The cost price of 20 articles is the same as the selling price of x articles. If the profit is 25%, then the value of x is:**

   **A.15** **B.16** **C.18** **D.25**

5. **Vinay runs 1000 meters in 20 seconds and ajay runs the sam distance in 25 seconds. By what distance will vinay beat Ajay in a 100 metre race ?**

   **A.10 m**
   **B. 12 m**
   **C. 20 m**
   **D. 24 m**

6. **Lucknow, Patna, Bhopal, Jaipur, ? ? ? ?**

   **A. Mysore**
   **B. Pune**
   **C. Shimla**
   **D. Indore**

7. **In how many different ways canthe letters of the word 'RETAI' be arranged in such a way that the vowels occupy only the odd positions ?**

## 8. Predict the output:

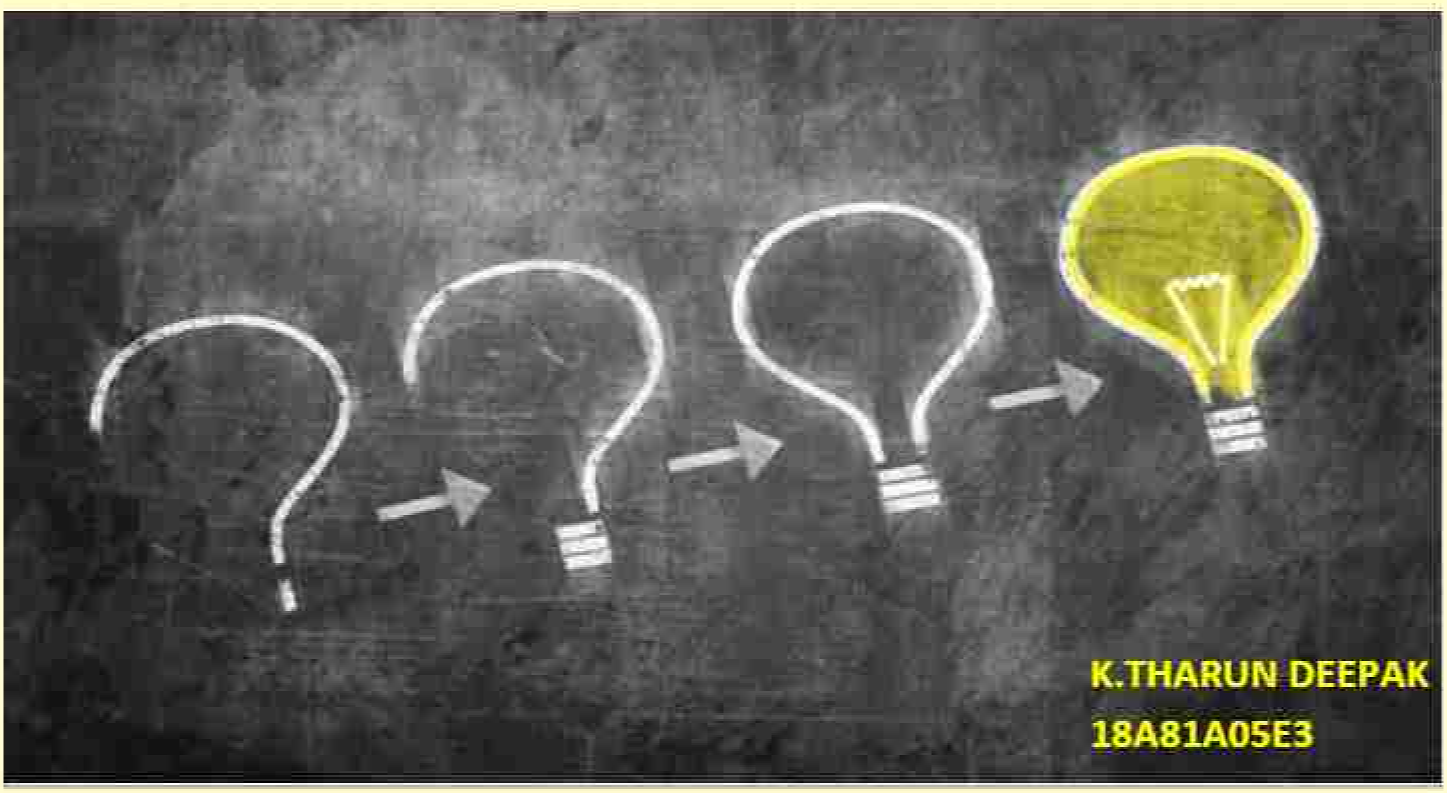
```
#include<iostream>
using namespace std;
int max(int& x,int& y,int& z)
{
  if (x>y && y>z)
  {
    y++;
    z++;
    return x++;
  }
  else
  {
    if (y>x)
      return y++;
    else
      return z++;
   }
}
 int main()
{
  int A,B, a=10,b=13,c=8;
  A=max(a,b,c);
  cout<<a++<<" "<<b--<<" "<<++c;
  B=max(a,b,c);
  cout<<++A<<" "<<--B<<" "<<c++;
  return 0;
 }
```

## 9. Predict the output:

```
#include<stdio.h>
int main()
{
 int i=20 , j ;
 i=(printf("Hello"));
 printf("%d",i);
 return 0;
 }
```

## 10. Find if any error in the code

```
main()
{
 int i=1;
 for(; ;)
 {
  printf("%d", i++);
  if(i>10)
  break;
  }
 }
```




V. Bhanu Teja
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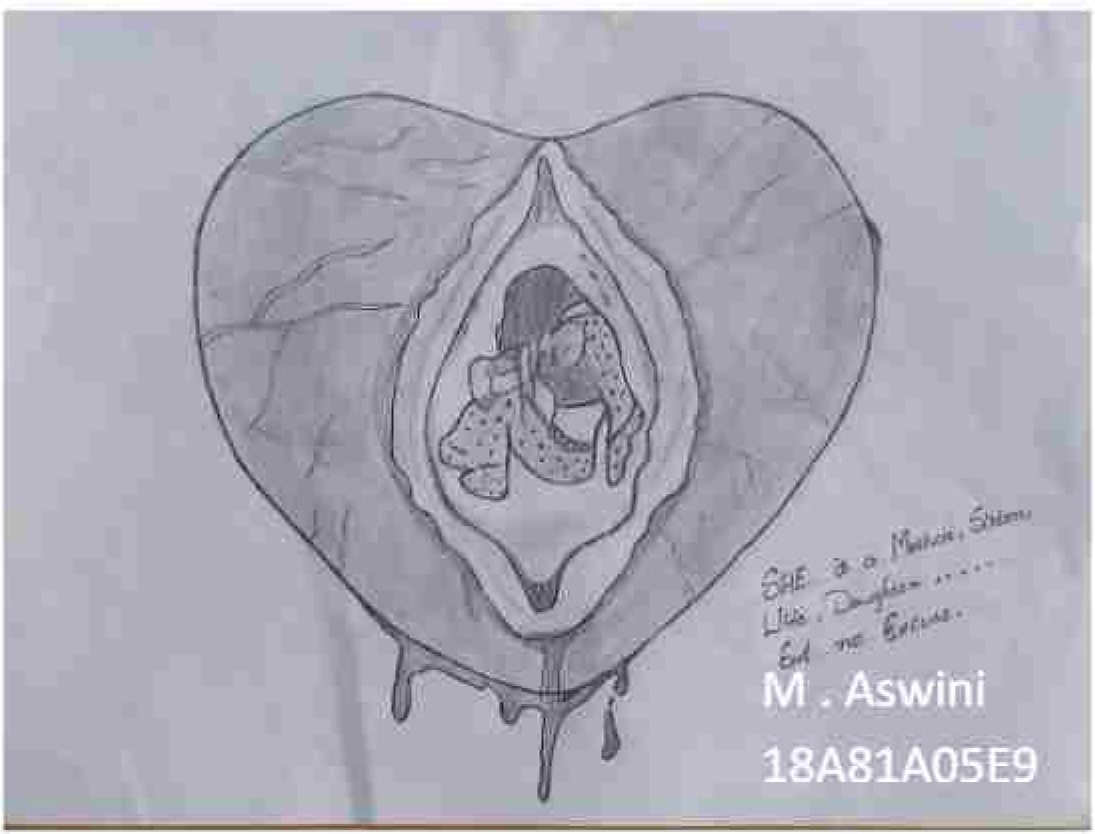
M . Aswini
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A Kamesh
18A81A05C1



K . Anjani
18A81A05K2


Web

Development with

**python**